作业 | 暂时保存 | 提交

第一章作业

题量: 10 满分: 100 作答时间: 09-22 13:22 至 10-08 13:22

一. 简答题（共10题，100分）

1. (简答题)
Five routers are to be connected in a point-to-point subnet . Between each pair of routers, the designers may put a high-speed line, a medium-speed line, a low-speed line, or no line. If it takes 100 ms of computer time to generate and inspect each topology, how long will it take to inspect all of them?

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2. (简答题)
What are two reasons for using layered protocols? What is one possible disadvantage of using layered protocols?

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3. (简答题)
What is the principal difference between connectionless communication and connection-oriented communication? Give one example of a protocol that uses
(1) connectionless communication
(2)connection-oriented communication

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4. (简答题)
In some networks, the data link layer handles transmission errors by requesting that damaged frames be retransmitted. If the probability of a frame's being damaged is p, what is the mean number of transmissions required to send a frame? Assume that acknowledgements are never lost.

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一. 简答题（100分）

1 2 3
6 7 8

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1 2 3

6 7 8

5. (简答题)
A system has an n-layer protocol hierarchy. Applications generate messages of length M bytes. At each of the layers, an h-byte header is added. What fraction of the network bandwidth is filled with headers?

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6. (简答题)
What is the main difference between TCP and UDP?

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7. (简答题)
When a file is transferred between two computers, two acknowledgement strategies are possible. In the first one, the file is chopped up into packets, which are individually acknowledged by the receiver, but the file transfer as a whole is not acknowledged. In the second one, the packets are not acknowledged individually, but the entire file is acknowledged when it arrives. Discuss these two approaches.

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8. (简答题)
An image is 1024 X 768 pixels with 3 bytes/pixel. Assume the image is uncompressed. How long does it take to transmit it over a 56-kbps modem channel?
Over a 1-Mbps cable modem?
Over a 10-Mbps Ethernet?
Over 100-Mbps Ethernet?
Over gigabit Ethernet?

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一. 简答题（100分）

1 2 3

6 7 8

9. (简答题)

Suppose the algorithms used to implement the operations at layer k is changed. How does this impact operations at layers k − 1 and k + 1?

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10. (简答题)

Suppose there is a change in the service (set of operations) provided by layer k. How does this impact services at layers k-1 and k+1?

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